

Dibyaprakash Patnaik

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PROFESSIONAL SUMMARY

Results-driven Computer Science graduate pursuing MCA at PES University, Bengaluru, with hands-on experience building cloud-native full-stack applications using microservices design patterns. Skilled in developing and deploying decoupled frontend and backend services using React, FastAPI, and Node.js, integrating front-end, back-end, and middleware components, and persisting data in both NoSQL (MongoDB) and RDBMS (PostgreSQL) databases. Proficient in RESTful API development, containerization with Docker, CI/CD pipeline design, and cloud deployments. Strong foundation in DevOps methodologies, real-time communication systems, and security best practices including JWT authentication and role-based access control. Eager to contribute to enterprise-scale cloud solutions in a collaborative development environment.

TECHNICAL SKILLS

- Languages:** Python, JavaScript, SQL, C
- Frontend:** React 19, Angular (Fundamentals), HTML5, CSS3, Vite, Tailwind CSS, shadcn/ui, Framer Motion
- Backend & APIs:** FastAPI, Node.js, Express.js, REST API Design, SOA Principles, API Integration & Productionization
- Databases:** PostgreSQL, MongoDB, Firebase, SQLAlchemy ORM, Data Modeling, Data Pipelines (Fundamentals)
- Cloud & DevOps:** Azure Fundamentals, Docker, CI/CD (GitHub Actions), Vercel, Netlify, Render, Cloud Deployments
- Architecture & Patterns:** Microservices Design Patterns, REST/SOA Services, API Gateway Concepts, Service Decoupling
- Real-Time & Security:** Socket.IO, WebSockets, JWT Authentication, Role-Based Access Control (RBAC)
- AI / ML & LLMs:** PyTorch (Fundamentals), Scikit-learn, NLP Basics, LLM Integration (OpenAI API / LangChain), Prompt Engineering
- Tools & Platforms:** Git, GitHub, VS Code, Postman, Linux CLI

PROJECTS

StudentPulse — Cloud-Based Student Management Platform

React 19, Vite, Tailwind CSS, shadcn/ui, Framer Motion, Recharts, Firebase, FastAPI, PostgreSQL, JWT, Docker

- Architected a microservices-based student management platform with decoupled frontend and backend services communicating via RESTful APIs, persisting structured data in PostgreSQL (RDBMS).
- Implemented JWT-based authentication with role-based access control (RBAC) for secure multi-user access across admin and student roles, ensuring application-level security and session management.
- Built interactive analytics dashboards with Recharts for real-time academic tracking, attendance monitoring, and performance visualization across multiple student cohorts.
- Deployed containerized backend service on Render and frontend on Netlify with automated CI/CD pipeline via GitHub Actions for continuous integration and delivery.

NetFlow Simulator — Interactive TCP/IP Network Visualizer

React 19, Vite, Tailwind CSS, Framer Motion, JavaScript, MongoDB, Vercel

- Designed and developed a cloud-deployed interactive TCP/IP simulator with a custom-built visualization engine to demonstrate packet encapsulation and data flow across all OSI network layers.
- Integrated MongoDB (NoSQL) for persistent simulation data storage and user session management; deployed publicly on Vercel with automated CI/CD workflows.
- Implemented dynamic animations using Framer Motion to visually represent protocol headers, payload wrapping, and layer transitions in real time.

ConnectHub — Real-Time Enterprise Communication Platform

React, Tailwind CSS, Node.js, Express.js, MongoDB, Socket.IO, WebSockets

- Developed a real-time intranet messaging platform integrating front-end (React) with back-end (Node.js/Express) middleware components using REST APIs and WebSocket communication protocols.
- Engineered role-based access control with granular permissions and WebSocket-powered live updates for enterprise team collaboration, broadcast announcements, and centralized user management.
- Designed scalable NoSQL data models in MongoDB for message threading, user profiles, and organizational hierarchy with efficient query patterns.

EDUCATION

Master of Computer Applications (MCA)

2025 – 2027

PES University, Hoskerahalli, Bengaluru
Currently Pursuing — Focus on Software Engineering, Cloud Computing & Advanced Algorithms

B.Sc. Computer Science (Honours)

2022 – 2025

Smart Gurukul Degree College | Berhampur University, Odisha
CGPA: 7.3 | 68% | Distinction

CERTIFICATIONS

- 100 Days of Code: The Complete Python Pro Bootcamp** — Udemy | 57 Total Hours (Aug 2026)

CORE CS KNOWLEDGE

Data Structures & Algorithms | Object-Oriented Programming | DBMS | Operating Systems | Computer Networks | Software Engineering

LANGUAGES & INTERESTS

English (Proficient), Hindi (Native), Odia (Native) | Badminton, Cricket, Gym | Open-Source Exploration, Learning New Technologies